

Q1.

Asif and Bharveen play a zero-sum game. The game is represented by the following pay-off matrix for Asif.

		Bharveen		
		I	II	III
Asif	I	4	2	6
	II	2	10	1
	III	3	1	5

- (a) Explain why Asif should never play strategy III.

(1)

- (b) Asif plays strategy I with probability p_1 and strategy II with probability p_2 . The value of the game is v . Use the matrix to write down **three** inequalities in the form $v \leq ap_1 + bp_2$.

(3)

The probabilities p_1 and p_2 also satisfy the inequalities

$$p_1 + p_2 \leq 1, \quad p_1 \geq 0 \quad \text{and} \quad p_2 \geq 0.$$

The Simplex tableau for the linear programming problem which will give the optimal strategy for Asif can be written as:

P	v	p_1	p_2	r	s	t	u	
1	-1	0	0	0	0	0	0	0
0	0	1	1	1	0	0	0	1
0	1	-4	-2	0	1	0	0	0
0	1	-2	-10	0	0	1	0	0
0	1	-6	-1	0	0	0	1	0

- (c) (i) Apply **one** iteration to this Simplex tableau.

(4)

- (ii) Explain why your answer to part (c)(i) is **not** optimal.

(1)

- (iii) After two further iterations, Asif's tableau is:

P	v	p_1	p_2	r	s	t	u	

5	0	0	0	18	4	1	0	18
0	0	0	10	2	1	-1	0	2
0	0	10	0	8	-1	1	0	8
0	5	0	0	18	4	1	0	18
0	0	0	0	14	-13	3	10	14

Show that the value of the game in this case is $3\frac{3}{5}$ and find Asif's optimal mixed strategy.

(3)

- (d) Assuming the value of the game is $3\frac{3}{5}$ and by considering mixed strategies, find the optimal mixed strategy for Bharveen.

(6)

(Total 18 marks)



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